

Windows Fundamentals 1

A complete, beginner-friendly guide to the Windows desktop, file system, user accounts and security basics.

Learning Path: Pre Security / Cyber Security 101 - Windows Fundamentals

Difficulty: Info | Topics: Windows editions, GUI, NTFS, user accounts, UAC, Task Manager

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About this walkthrough

This document walks through every task of the TryHackMe room **Windows Fundamentals 1**. Each task explains the concept being taught, states the exact question asked, describes how to reach the answer in the deployed Windows VM, and gives the answer itself.

What you will learn

- How Windows editions (Home vs Pro) differ and what BitLocker adds.
- How to navigate the Windows desktop, taskbar and notification area.
- How the NTFS file system stores and protects data.
- How user accounts, groups, permissions and UAC keep Windows secure.

Contents

Task 1	Introduction to Windows	3
Task 2	Windows Editions	4
Task 3	The Desktop (GUI)	5
Task 4	The File System	6
Task 5	The Windows\System32 Folders	7
Task 6	User Accounts, Profiles and Permissions	8
Task 7	User Account Control	9
Task 8	Settings and the Control Panel	10
Task 9	Task Manager	11
Task 10	Conclusion	12

Before you start

Deploy the Windows virtual machine with the green **Start Machine** button at the top of the room. Try each task yourself before reading the answers - the whole point of this room is to get comfortable clicking around Windows. Some answers depend on the state of the deployed VM (for example the last item in the Control Panel list), so yours may differ slightly if the machine image is updated. All answers are shown for educational purposes.

TASK 1

Introduction to Windows

Windows is the world's most widely used desktop operating system - roughly three out of every four desktop and laptop computers run it. Its popularity comes from an easy-to-learn graphical interface, broad hardware support and the fact that most office and gaming software is written for it. That same market share is exactly why attackers target it so heavily, and why a security professional needs to understand its internals.

This room is the first of three parts. In part 1 you explore the parts of Windows a normal user sees every day: the desktop, the file system, user accounts, the Control Panel and the Task Manager. Later parts cover system tools, Windows Security, updates, and the command line.

The TryHackMe VM

This room starts a **Windows** virtual machine using the green **Start Machine** button - it is separate from the Linux AttackBox. Once it is running, open the remote desktop in your browser and use it for every interactive question in this walkthrough.

Q1. Read above and start the virtual machine.**ANSWER**

No answer needed - just start the machine

TASK 2

Windows Editions

Microsoft has released many versions of Windows since 1985, and each version comes in different **editions** aimed at different users. Two editions you will meet constantly are **Home** (for ordinary consumers) and **Pro** (for businesses and power users).

Capability	Home	Pro
Everyday use, apps, games, browsing	Yes	Yes
Join a corporate domain / Azure AD	No	Yes
Group Policy management	No	Yes
Remote Desktop (host)	No	Yes
BitLocker disk encryption	No	Yes

BitLocker is a full-disk encryption feature. When it is enabled, the entire drive is encrypted and protected by a key, so that if a laptop is lost or stolen the data cannot simply be read by removing the disk. This makes it a genuinely important security control - and one of the clearest reasons organisations choose Windows Pro over Home.

Q1. What encryption can you enable on Pro that you can't enable in Home?

ANSWER

BitLocker

TASK 3

The Desktop (GUI)

The **Desktop** is the graphical interface you land on after logging in. Under the hood it is really an aggregated view of two folders: a public desktop (C:\Users\Public\Desktop) that every user sees, and a per-user desktop (%userprofile%\Desktop). Creating a shortcut is often just dropping a file into one of those folders.

The strip along the bottom of the screen is the **taskbar**, and it contains several components worth knowing:

- **Start Menu** - menu-based access to applications, settings and power options.
- **Search Box** - type here to find apps and files quickly; you can hide it, show it as an icon, or show it in full.
- **Task View** - a bird's-eye view of all open windows and virtual desktops; it can be shown or hidden.
- **Toolbars** - small extra toolbars (for example Desktop) that add quick navigation.
- **Notification Area** - the clock, network, volume, battery and status icons such as the Action Center.

How to change taskbar options

Right-click an empty part of the taskbar and choose **Taskbar settings**. There you can toggle the Search box style and the Task View button. Right-clicking the notification icons opens the **Action Center** (now called the Notification Center).

Q1. Which selection will hide/disable the Search box?

ANSWER

Hidden

Q2. Which selection will hide/disable the Task View button?

ANSWER

Show Task View button

Q3. Besides Clock, Volume, and Network, what other icon is visible in the Notification Area?

ANSWER

Action Center

TASK 4

The File System

A **file system** is the standardized way a storage device stores and retrieves data. Early versions of Windows used **FAT** (File Allocation Table) - FAT16 and FAT32 - but those had strict limits (most famously a maximum file size of 4 GB). Modern Windows uses **NTFS**.

NTFS is a **journaling** file system: it keeps a log of changes so that, after a crash or power failure, damaged files and folders can be repaired consistently. It also adds features that matter for security:

- Files larger than 4 GB are supported.
- Per-file and per-folder **permissions** (full control, modify, read & execute, list folder contents, read, write) can be set for specific users and groups.
- Folders and files can be **compressed** to save space.
- **Alternate Data Streams (ADS)** let a single file hold more than one data stream - useful for metadata, but also abused by attackers to hide data.

Q1. What is the meaning of NTFS?

ANSWER

New Technology File System

TASK 5

The Windows\System32 Folders

The operating system itself lives in the **Windows** folder, normally C:\Windows. Windows does not assume that path, though - it stores it in an environment variable so the OS still works if it is installed elsewhere. The system variable that points at the Windows folder is written **%windir%**, and you can type it straight into File Explorer's address bar to jump there.

Inside that folder sits **System32**. It contains the core, critical files that Windows needs to run - device drivers, DLLs and many of the built-in executables. Because it is so important, you should be very careful in this directory: deleting or editing the wrong file can stop Windows from booting.

```
%windir%           -> resolves to C:\Windows  
C:\Windows\System32 -> core OS files (treat as read-only)
```

Q1. What is the system variable for the Windows folder?

ANSWER

%windir%

TASK 6

User Accounts, Profiles and Permissions

Windows has two broad kinds of account. An **Administrator** can change system settings, install software, and add or delete users and groups. A **Standard User** can only change the parts of the system that they have been given access to, and cannot make system-wide changes.

A new user gets a **profile** - a personal set of folders (Desktop, Documents, Downloads, Music, Pictures and so on) created for them. Accounts can be managed from Settings (type Other Users in the search box) or, for finer control, from **Local User and Group Management**.

```
lusrmgr.msc
Users    -> every local account and its status
Groups  -> Administrators, Users, Remote Desktop Users, ...
A user inherits the permissions of every group they belong to.
```

Run `lusrmgr.msc`, open the **Users** folder and look at the accounts. Alongside the built-in **Administrator** there is an extra account, plus the built-in **Guest** account, whose description is "Built-in account for guest access to the computer/domain". Open a user's **Properties** and the **Member Of** tab to see which groups they belong to, and the **General** tab to see whether the account is enabled or disabled.

Q1. What is the name of the other user account?

ANSWER

tryhackmebilly

Q2. What groups is this user a member of?

ANSWER

Remote Desktop Users,Users

Q3. What built-in account is for guest access to the computer?

ANSWER

Guest

Q4. What is the account status?

ANSWER

window\$Fun1!

TASK 7

User Account Control

Running every day as a full administrator is risky: a single mistake or piece of malware then has complete control of the machine. Linux solves this with `sudo` - you stay a normal user and only raise your privileges for specific commands. Windows has an equivalent mechanism called **User Account Control (UAC)**.

With UAC, even a user who is in the Administrators group runs with standard privileges by default. When something genuinely needs administrator rights, Windows asks for confirmation - and depending on the setting, may require an administrator password - before the action is allowed. This stops silent, unauthorised changes and is a key part of Windows' defence-in-depth.

Q1. What does UAC mean?

ANSWER

User Account Control

TASK 8

Settings and the Control Panel

Windows gives you two main places to change how the system behaves. The modern **Settings** app covers the common, day-to-day options, while the older **Control Panel** exposes deeper and more technical settings. In daily use you will often bounce between the two, and the fastest way to reach either is to type its name (or a keyword) into the search box next to the Start Menu.

The Control Panel can display its items in several ways. Use the **View by** drop-down in the top-right corner to switch between Category and large/small icon views. Choosing **Small icons** lists every applet alphabetically, and the final entry in that list is what this task asks for.

Why your answer might differ

The last item in the Control Panel depends on which applets are installed on the machine. On the TryHackMe Windows VM it is **Windows Defender Firewall**; on a personal PC it might be something like Windows Mobility Center instead.

Q1. In the Control Panel, change the view to Small icons. What is the last setting in the Control Panel view?

ANSWER

Windows Defender Firewall

TASK 9

Task Manager

The **Task Manager** is one of the most-used utilities in Windows. It lists every running process and lets you inspect and act on them - most importantly, ending a process that has frozen. It also shows how much CPU, memory, disk and network each process is using, which makes it a first stop when a machine is slow or when you suspect something unwanted is running in the background.

Method	Result
Ctrl + Shift + Esc	Opens Task Manager directly
Ctrl + Alt + Del, then Task Manager	Opens the security options screen first
Right-click the taskbar, then Task Manager	Opens it from the context menu

Q1. What is the keyboard shortcut to open Task Manager?

ANSWER

Ctrl+Shift+Esc

TASK 10

Conclusion

This room introduced the Windows operating system from a user's point of view: the editions Microsoft sells (and the BitLocker encryption that only Pro offers), the desktop and taskbar, the NTFS file system that stores your data, the Windows and System32 system folders, user accounts, profiles and groups, the protective role of User Account Control, and the everyday control surfaces - Settings, the Control Panel and the Task Manager.

Terminate the machine

Once you have finished the room, remember to stop the Windows virtual machine from the room page so it does not keep running in the background.

Answer key at a glance

Every question in the room, in one place. Use this to check your own answers only after you have worked through the tasks in the deployed machine.

Task	Question	Answer
1	Start the virtual machine	No answer needed
2	Encryption on Pro but not Home	BitLocker
3	Selection that hides the Search box	Hidden
3	Selection that hides the Task View button	Show Task View button
3	Extra icon in the Notification Area	Action Center
4	Meaning of NTFS	New Technology File System
5	System variable for the Windows folder	%windir%
6	Name of the other user account	tryhackmebilly
6	Groups that user belongs to	Remote Desktop Users,Users
6	Built-in account for guest access	Guest
6	Account status	Account is disabled
7	Meaning of UAC	User Account Control
8	Last setting in the Control Panel	Windows Defender Firewall
9	Keyboard shortcut to open Task Manager	Ctrl+Shift+Esc
10	Terminate the machine	No answer needed

Next steps

Continue with **Windows Fundamentals 2**, which explores the System Configuration utility, advanced system settings, the registry, the Resource Monitor, and other built-in admin tools. After that, **Windows Fundamentals 3** covers Windows Update, Windows Security and the other built-in protections.